

UNIVERSITY EXAMINATION BOARD
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
SEMESTER FINAL DAA EXAM

Subject: Design and Analysis of Algorithms	Sub Code: CS-501	Max Marks: 50
Time / Duration: 3 Hours	Academic Year: 2026-2027	Semester: V Semester

General Instructions:

- 1. Answer all questions.
- 2. Assume suitable data if necessary.

PART A

1. Critically evaluate Regular Expressions against alternative paradigms. When is Regular Expressions strictly optimal? **[10 Marks]**
1. Discuss the trade-offs in time, space, and scalability when engineering systems centered around LR Parsers. **[10 Marks]**
1. Discuss the trade-offs in time, space, and scalability when engineering systems centered around Three Address Code. **[10 Marks]**
1. Derive and explain the mathematical recurrence or governing equations of Flow Graphs. Prove its correctness. **[10 Marks]**
1. Design a robust solution using DAG Representation. for an enterprise-scale engineering problem. Analyze its complexity. **[10 Marks]**